



Explorative Data Analysis

Leveraging Data Analytics for Efficient OR Planning at ZOL

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Introduction

This document provides an exploratory overview of the dataset used for analyzing operating room (OR) efficiency at Ziekenhuis Oost-Limburg. The aim is to describe the main temporal, clinical, and operational characteristics of the data before moving toward more specific performance metrics. By systematically reviewing the distributions, frequencies, and summary statistics of key variables, we gain a first impression of how surgical activity is structured across time, departments, and procedure types.

The analysis follows a descriptive approach, highlighting patterns that may explain later deviations between planned and observed activity. Each visualization or table is accompanied by a short interpretation that connects numerical trends to the daily practice of the operating theatre. Together, these insights form the empirical foundation for subsequent modelling and the development of operational key performance indicators (KPIs) focused on scheduling accuracy, resource allocation, and overtime work.

Different sites

The distribution of procedures across the different campuses reflects the operational profile of Ziekenhuis Oost Limburg. As expected, the largest share is concentrated at the main campus in Genk, which houses the largest number of operating rooms and accommodates a wide range of surgical specialties. Maaseik and Lanaken contribute smaller but still substantial volumes, with Lanaken focusing primarily on day surgery. The Cathlab accounts for a relatively small proportion of the activity, yet it remains important to include it in the analysis as procedures performed there also rely on operating room capacity and anesthesiology support. This variation in case distribution underlines the need to account for campus specific characteristics when assessing overall efficiency and planning capacity. Figure 1 shows the relative shares across the four locations.

Relative share of surgical procedures per campus

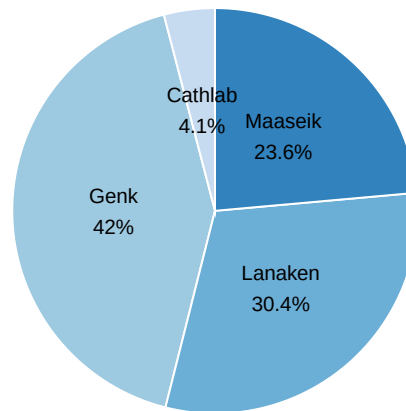


Figure 1: Relative share of surgical procedures per campus

Descriptive analysis

The descriptive analysis provides a structured overview of the main variables of the dataset before moving to more advanced comparisons between planning and execution. It summarizes how surgical activity is distributed across time, procedure types, and operational variables, offering a factual foundation for interpreting later results. By examining these baseline distributions, we can better understand the context in which deviations, overtime, and scheduling inefficiencies occur.

Temporal information

Understanding when surgeries take place is essential for interpreting overall operating room performance. Temporal variables provide insight into how surgical activity is distributed across years, weekdays, and hours of the day. These patterns reflect both organizational planning and patient demand, and they influence key operational factors such as staff allocation, overtime work, and resource utilization. The following subsections explore these temporal dimensions to establish a baseline view of how surgical activity is structured over time.

Year

Surgical activity remained relatively stable over the four year period, with a gradual increase from 2022 (27.5 percent) to 2024 (30.4 percent). This upward trend suggests a steady expansion in operating room capacity and case throughput. The lower proportion observed in 2025 (12.9 percent) is expected, as the dataset only includes partial data for that year. Table 1 provides the yearly counts and percentages.

Table 1: Number of surgeries per year (count and percentage)

Year	n	Percentage
2022	70599	27.5%
2023	75135	29.2%
2024	78053	30.4%
2025	33136	12.9%

Weekday

Surgical activity is clearly concentrated during regular weekdays, with each weekday from Monday to Friday accounting for roughly one fifth of all procedures. The distribution is balanced, ranging from 19.0 percent on Wednesdays to 20.7 percent on Tuesdays. Weekend activity is minimal, with 0.9 percent on Saturdays and 0.8 percent on Sundays, which aligns with the hospital’s scheduling policy that reserves weekend days mainly for urgent or emergency cases. Table 2 provides the detailed counts and percentages.

Table 2: Number of surgeries per weekday (count and percentage)

Weekday	Count	Percentage
Di	53130	20.7%
Do	49319	19.2%
Ma	50072	19.5%
Vr	51328	20.0%
Wo	48793	19.0%
Za	2267	0.9%

Weekday	Count	Percentage
Zo	2014	0.8%

StartHour

Most surgeries start between 08:00 and 15:00, which aligns with regular operating room hours. The peak at 08:00 reflects the beginning of the daytime shift, when most planned cases commence. A gradual decline is observed throughout the afternoon, with only a small number of procedures starting after 16:00. Evening and night time activity remains limited, indicating that non elective or emergency surgeries are relatively infrequent compared to daytime operations. This clear concentration of start times confirms that the surgical workflow at Ziekenhuis Oost Limburg is primarily organized around standard daytime hours. Figure 2 shows the full distribution of start hours.

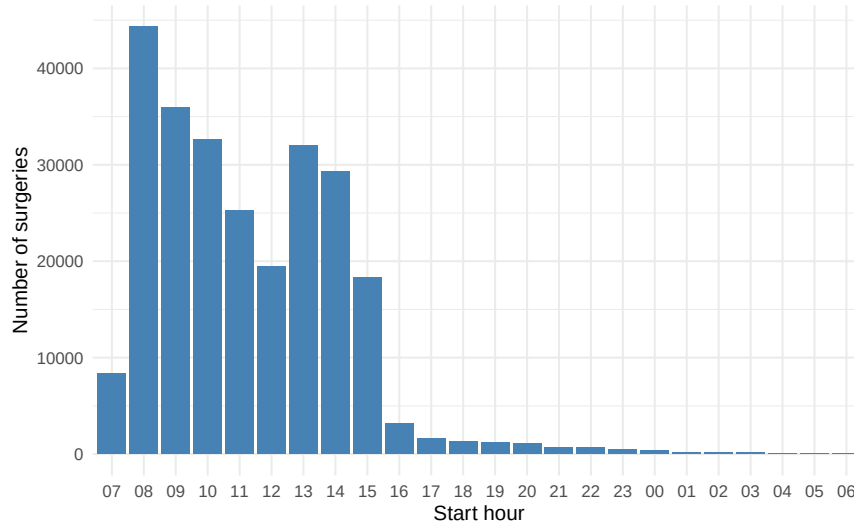


Figure 2: Distribution of surgery start times

Clinical characteristics

Clinical characteristics describe the type of surgical activity captured in the dataset. These variables outline what procedures were performed, under which admission conditions, and with what urgency level. Together, they provide an overview of the clinical composition of the surgical workload and highlight how case types and patient pathways influence planning complexity. Understanding this context is essential before interpreting operational performance or scheduling outcomes.

ProcedureName

The dataset includes a total of 2222 unique procedure names. On average, each procedure type was performed 115.2 times, while the median frequency equals 13 cases. This large difference between the mean and the median confirms that surgical activity is highly concentrated in a small number of frequently performed procedures, with many others occurring only occasionally. Table 3 provides an overview of these summary statistics.

Table 3: Overview of surgical procedures

Unique procedures	Average cases per procedure	Median cases per procedure
2222	115.2	13

The twenty most frequent procedure names together represent a substantial share of all surgeries. The most common procedure exceeds 12000 cases, while those near the bottom of the top twenty occur between 2000 and 4000 times. This shows a steep concentration in activity, where a small set of procedures dominates the dataset. The difference between the upper and lower ranks confirms that most surgical volume is generated by a limited number of recurring procedures, while many others appear only a few times across the study period. Figure 3 displays the twenty most frequently recorded procedure names.

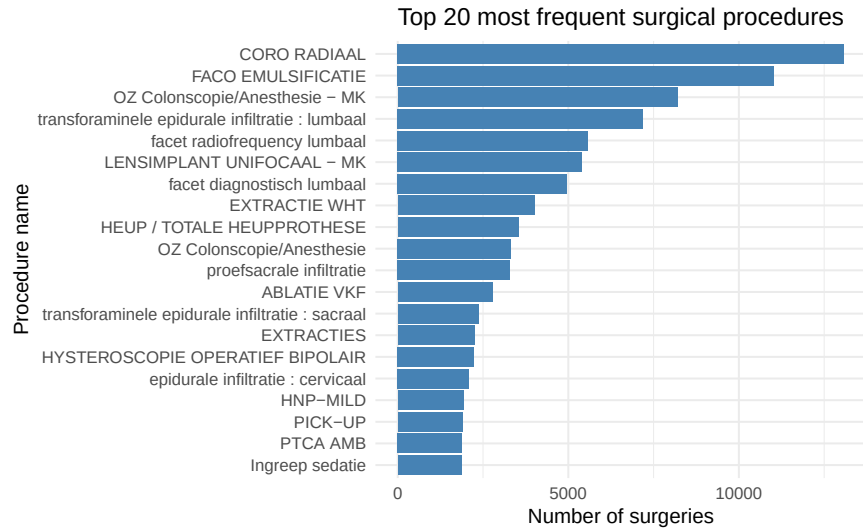


Figure 3: Top 20 most frequent surgical procedures

AdmissionType

Most surgeries in the dataset are registered as ambulant cases, representing 181948 procedures or 70.4 percent of all activity. Hospital admissions account for 76371 procedures, or 29.5 percent. Only a very small share, 198 procedures or 0.1 percent, concerns emergency cases. This distribution confirms that the majority of surgical activity takes place in a planned short stay context, while inpatient and emergency procedures form a much smaller part of the total workload. The clear dominance of day surgery also reflects the increasing shift toward efficient same day interventions within the hospital's operating schedule. Table 4 provides the detailed counts and percentages.

Table 4: Number of surgeries per admission type (count and percentage)

AdmissionType	Count	Percentage
DAG	180932	70.4%
HOS	75800	29.5%
SPOED	191	0.1%

UrgencyType

The vast majority of surgeries are classified as elective, representing 240401 procedures or 93.6 percent of all cases. Non elective cases account for 16522 surgeries, or 6.4 percent. This limited share of urgent activity suggests that most procedures are scheduled in advance and follow a predictable planning process. Non elective cases, although relatively few, can still have a significant operational impact since they often require last minute scheduling and may lead to deviations in the planned operating room program. Table 5 provides the detailed counts and percentages.

Table 5: Number of surgeries per urgency type (count and percentage)

UrgencyType	Count	Percentage
electief	240401	93.6%
niet-electief	16522	6.4%

Hospital stay

Hospital stay variables capture how long patients remain admitted after surgery. Analyzing these distributions helps identify whether short- or longstay cases dominate the dataset and how different admission or urgency types affect discharge timing. The results in this section provide a bridge between surgical scheduling and patient flow within the hospital.

LengthOfStay

The average hospital stay after surgery is 2.974 days, with a median of 1 day, indicating that most patients are discharged quickly after their procedure. The distribution is highly skewed, as shown by the large difference between the mean and the median. While 95 percent of stays are shorter than 9 days, a few exceptional cases extend far longer, up to 953 days. These long stays represent rare outliers and likely reflect complex or rehabilitation related admissions rather than typical surgical recoveries. Table 6 summarizes the main summary statistics for length of stay.

Table 6: Length of stay, summary statistics

n	Mean	Median	SD	IQR	Min	Max	P95	P99	P99_9
256898	2.974	1	9.481	1	1	953	9	38.03	134

The distribution of hospital stay without revalidation confirms that most patients are discharged within a single day. The frequency drops sharply after the first day, and only a small number of cases remain hospitalized for more than three days. This right skewed pattern illustrates that short stays dominate the dataset, while extended admissions are relatively uncommon. Figure 4 shows the truncated distribution up to the 96th percentile.

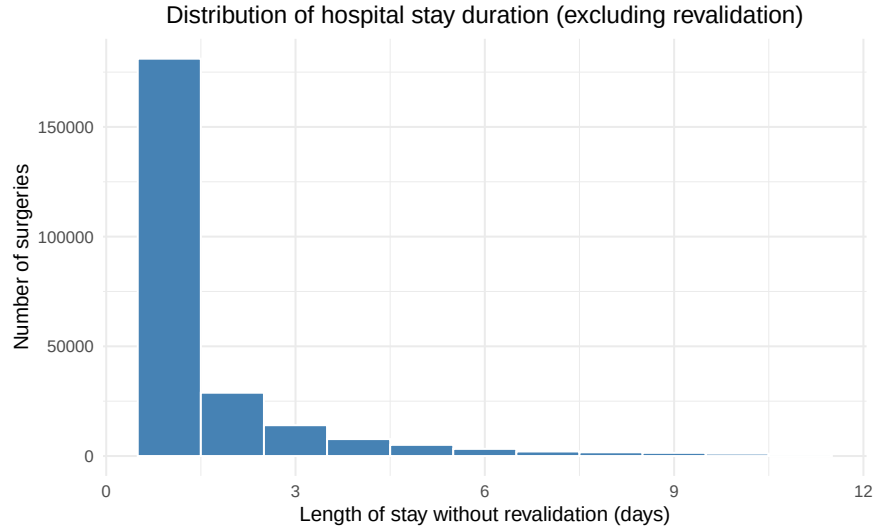


Figure 4: Distribution of hospital stay duration without revalidation

When comparing length of stay across admission and urgency types, clear differences become visible. Day cases have consistently short stays, regardless of urgency, while hospitalized patients show a much wider spread. Non elective hospital admissions tend to have longer and more variable stays than elective ones, as shown by the wider and higher distributions. Emergency cases appear in very limited numbers and follow a similar pattern of variability. Overall, these distributions confirm that length of stay is strongly influenced by both the admission type and the urgency of the surgery. Figure 5 provides an overview of these differences.

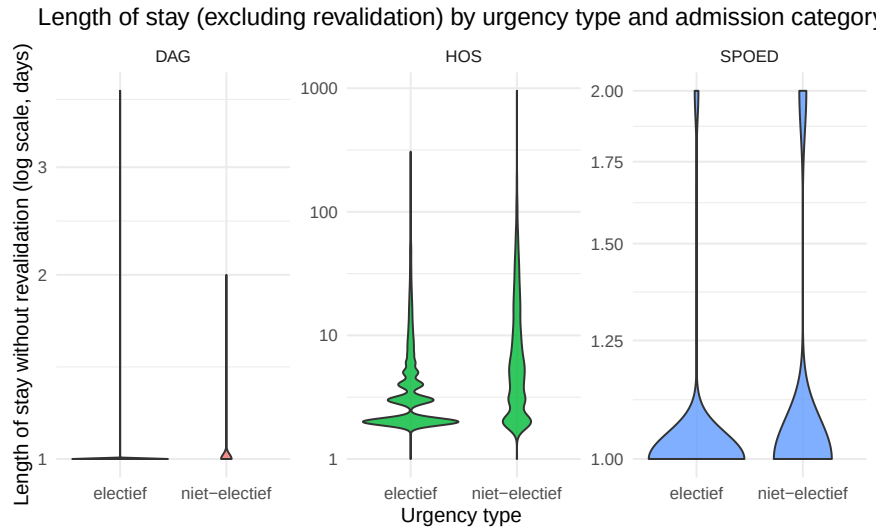


Figure 5: Length of stay without revalidation by urgency type and admission category

Planned surgery information

This section summarises all variables related to the planned part of the surgical workflow. It describes how surgeries were initially scheduled in terms of expected duration, operating room assignment, and distribution across rooms. Examining these metrics allows us to understand how operating capacity is allocated in advance

and to what extent planning patterns differ between rooms. These insights serve as a reference point for comparing planned versus actual execution later in the report.

PlannedDurationMinutes

The planned duration of surgeries varies widely across the dataset. On average, procedures are scheduled for 56.53 minutes, while the median is 35 minutes, indicating that most surgeries are relatively short but a smaller group of long procedures increases the mean. The interquartile range of 53 minutes shows substantial variation within the middle 50 percent of cases. Nearly all surgeries are planned to last less than 150 minutes, and only a very small number exceed 317 minutes. The 99.9th percentile equals 854.1 minutes, representing rare outliers with exceptionally long planned durations. This distribution suggests that the majority of planned surgical activity consists of short to medium length procedures. Table 7 summarises these statistics.

Table 7: Planned duration of surgeries in minutes, summary statistics

Mean	Median	SD	IQR	Min	Max	P95	P99	P99.9
56.53	35	67.81	53	1	1439	150	317	854.1

The distribution of planned surgery durations shows a strong right skewed pattern. Most procedures are scheduled for less than 60 minutes, with a clear peak between 20 and 50 minutes. Beyond that point, the frequency gradually declines, and only a small fraction of surgeries are planned to last more than two hours. This indicates that short and medium length procedures dominate the planned surgical workload, while long operations represent a relatively small portion of the overall schedule. Figure 6 shows the trimmed distribution up to the 99th percentile.

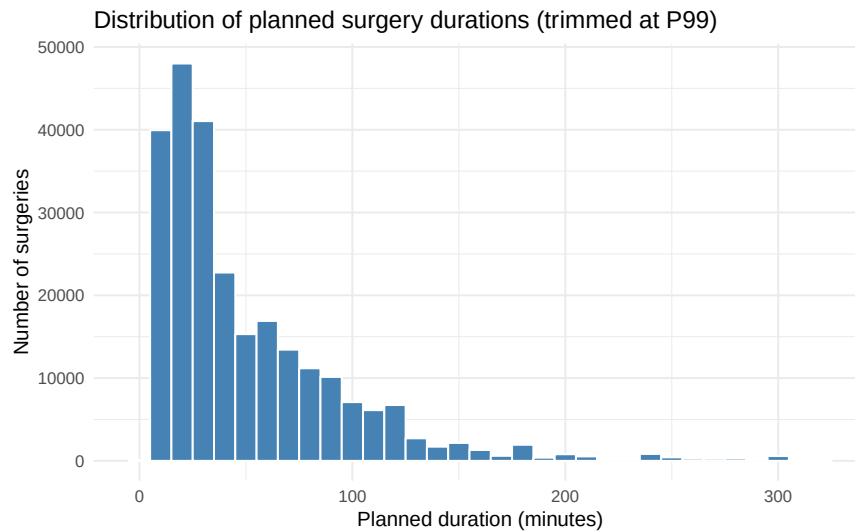


Figure 6: Distribution of planned surgery durations (trimmed at P99)

PlannedOR

The table below provides a detailed overview of surgical planning across all operating rooms. For each room, it lists the number of planned surgeries, summary statistics of planned durations, and the number of unique procedure names. This gives a comprehensive view of both workload and diversity in the planned surgical program. Table 8 summarises these metrics.

Planned activity is unevenly distributed across rooms. The most active rooms, LP02 and LP01, together represent 15.1 percent of all planned surgeries, with 19868 and 18940 cases respectively. In contrast, a large number of rooms handle only a few hundred cases per year. Average planned durations vary considerably between rooms, ranging from 14.4 to 286.7 minutes. This variation indicates that certain rooms are used for shorter high throughput procedures, while others are scheduled for longer or more complex cases.

The coefficient of variation, shown in Table 8, ranges from 0.18 to 2.26, which demonstrates that variability within rooms is substantial. Procedure diversity also differs strongly between rooms. Some have only a handful of recurring procedure types, while others register several hundred unique procedure names. Together, these patterns highlight that surgical planning at Ziekenhuis Oost Limburg is far from uniform across operating rooms, with clear differences in both workload and procedural variety.

Table 8: Per operating room: number of planned surgeries, planned duration metrics (minutes), and procedure diversity

PlannedOR	Count	Percentage	Mean	Median	SD	CV	IQR	P95	UniqueProcedures
LP02	19868	7.7%	17.1	15	7.5	0.44	5	26	108
LP01	18940	7.4%	14.4	15	4.7	0.32	10	20	52
MKI3	14163	5.5%	18.2	15	7.7	0.42	5	32	179
LO02	10018	3.9%	33.3	29	17.4	0.52	18	60	100
KO04	9255	3.6%	36.9	27	73.3	1.99	10	50	37
KO03	8845	3.4%	54.3	30	122.6	2.26	19	83	51
MO02	8301	3.2%	26.7	24	12.5	0.47	6	45	116
MKI1	8245	3.2%	16.6	17	9	0.54	4	19	6
GO11	7534	2.9%	77	67	51.8	0.67	50	154	615
GO14	6894	2.7%	51.6	40	50.7	0.98	47	120	230
GO01	6618	2.6%	61.7	36	82.3	1.33	30	178.1	198
LO04	5898	2.3%	44.2	32	25	0.57	26	105	136
MO04	5549	2.2%	69.9	74	35.2	0.5	48	120	421
GO17	5496	2.1%	75.6	78	48.6	0.64	60	147	274
GO18	5298	2.1%	74.9	72	53.5	0.71	49	150	280
GO06	5232	2.0%	86.4	48	90.8	1.05	52	248.3	164
LO05	5229	2.0%	54.8	45	41.6	0.76	50	111	223
LO01	5217	2.0%	54.6	51	24.3	0.45	26	98	253
GO05	4888	1.9%	103.8	73	90	0.87	68	245	298
KO06	4839	1.9%	58.8	52	72.7	1.23	27	113	56
GO07	4754	1.9%	77.3	67	60.4	0.78	40	160	313
MO03	4542	1.8%	71.6	64	55.8	0.78	42	185	382
GO04	4519	1.8%	104.6	72	92.4	0.88	63.5	289	264
MO06	4508	1.8%	50.2	45	31.9	0.63	29	120	326
GEG1	4489	1.7%	24.5	21	14.4	0.59	15	48	23
MO05	4465	1.7%	70.8	70	37.5	0.53	55	120	407
GO15	4314	1.7%	89.1	80	54.2	0.61	68.8	164	323
LO03	4222	1.6%	68.4	60	41	0.6	61	121	273
GO03	4207	1.6%	98.6	81	69.6	0.71	58	200	189
KO01	4126	1.6%	83.3	75	71.3	0.86	42	125	43
GO16	4096	1.6%	95.3	101	45.4	0.48	50	150	284
MO01	4049	1.6%	63.4	60	35.5	0.56	55	120	271
KO02	3880	1.5%	86.3	86	40.5	0.47	51	123	31
KO07	3516	1.4%	78.1	65	72.8	0.93	77	159	74
GO02	3510	1.4%	109.2	80	93.7	0.86	83	276.3	242
GO13	3291	1.3%	138.8	110	94.7	0.68	74	305	189
KO05	2997	1.2%	85.7	76	67.8	0.79	23	144	77
GO12	2909	1.1%	144.9	110	95.3	0.66	91	320.6	161

PlannedOR	Count	Percentage	Mean	Median	SD	CV	IQR	P95	UniqueProcedures
GOP1	2830	1.1%	35.3	39	22.5	0.64	12	52	50
GO09	2647	1.0%	144.6	124	102.7	0.71	123	322	264
GO08	2601	1.0%	92.9	60	75.1	0.81	65	255	156
GEE1	2338	0.9%	32.1	25	50.4	1.57	7	58	8
GSE1	2253	0.9%	28.7	30	16	0.56	10	45	17
GO10	1751	0.7%	286.7	318	146.2	0.51	179	480	178
GEE2	1558	0.6%	26.2	25	26	0.99	11	58	5
GOP2	1224	0.5%	33.7	30.5	14.1	0.42	13	50	64
GEX1	588	0.2%	21.7	20	6.2	0.28	0	25	4
ERCP	40	0.0%	44.1	46	8	0.18	9.2	52	5
MOR0	14	0.0%	25.4	20	15.8	0.62	7	61	4
GOR0	11	0.0%	63.4	61	56.6	0.89	27.5	149.5	8
LOR0	2	0.0%	39.5	39.5	34.6	0.88	24.5	61.5	2
GIC7	1	0.0%	75	75	NA	NA	0	75	1

The distribution of planned surgeries across operating rooms is highly uneven. A few rooms, notably LP02, LP01, and MKI3, account for a large share of total planned activity, each exceeding 10000 to 20000 cases. After these high volume rooms, the number of planned surgeries declines gradually, with most rooms showing between 2000 and 6000 scheduled procedures. Toward the right side of the chart, several rooms appear with very limited activity, often below 1000 cases. This steep decline illustrates the concentration of planned surgical volume in a small subset of operating rooms, while many others are used less intensively. Figure 7 shows the full distribution across rooms.

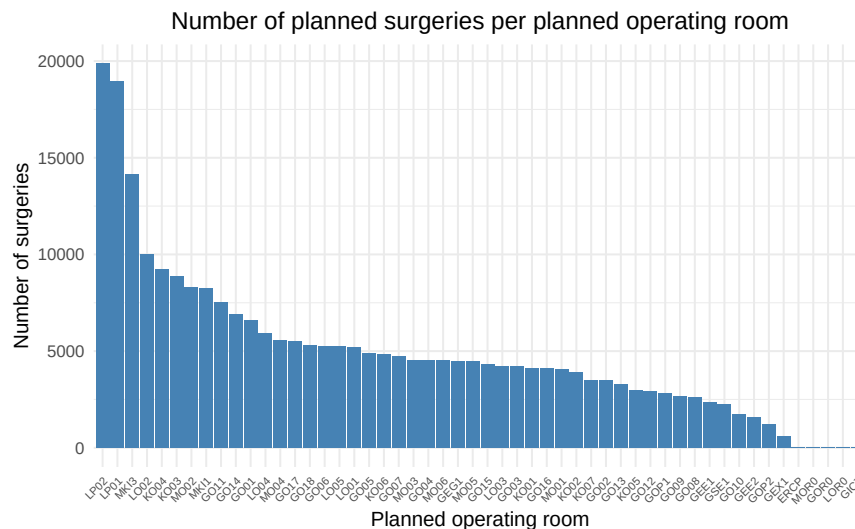


Figure 7: Number of planned surgeries per planned operating room

Actual surgery information

The actual surgery information provides an overview of how surgical activity was ultimately carried out in practice. By examining actual durations, operating room use, and volume distribution, this section highlights how the execution of surgeries compares to the initial plan. Understanding these real-world patterns is essential for interpreting deviations, overtime, and workflow efficiency in the following analyses.

ActualOR

The table below summarizes actual surgical activity per operating room, including the number of performed cases, average and median duration, variability measures, and the number of unique procedure names. It provides an overview of how operating room usage compares in terms of both workload and procedural diversity. Table 9 shows these results.

Similar to the planning data, surgical activity is concentrated in a small group of rooms. LP01, LP02, and MKI3 together account for more than 20 percent of all recorded surgeries, with 20189, 18619, and 14163 surgeries respectively. Most other rooms range between 2000 and 7000 surgeries, while a few have very low volumes below 1000. The distribution of mean durations is broad: several rooms have short procedures averaging between 14.8 and 30 minutes, while others show averages well above 100 minutes. This indicates substantial differences in typical case lengths.

The coefficient of variation, shown in Table 9, mostly ranges between 0.5 and 1.0, pointing to high internal variability in procedure duration within many rooms. The diversity of procedure types also differs strongly, from fewer than 10 unique procedures in some rooms to more than 600 in others. This large variation shows that some operating rooms handle highly specialized activities, while others are used for a wide mix of procedures. Overall, these results demonstrate that actual OR utilization, like the planning data, is heavily imbalanced across rooms, with notable heterogeneity in both volume and case complexity.

Table 9: Per operating room: number of actual surgeries, duration metrics (minutes), and procedure diversity

ActualOR	Count	Percent	Mean	Median	SD	CV	IQR	P95	UniqueProc
LP01	20189	7.9%	14.8	13	7	0.47	8	27	70
LP02	18619	7.2%	16	15	8	0.5	9	30	104
MKI3	14163	5.5%	15.9	13	10.1	0.64	10	36	180
LO02	9974	3.9%	33.2	28	17.1	0.51	21	66	101
KO04	9335	3.6%	30.8	27	15.7	0.51	15	61	36
KO03	8759	3.4%	39.9	33	27.8	0.7	26	88	53
MO02	8293	3.2%	26.9	24	13.6	0.51	10	49	114
MKI1	8262	3.2%	16.7	16	9.2	0.55	8	30	13
GO11	7567	2.9%	83.3	72	53.1	0.64	54	180	619
GO14	6896	2.7%	50.6	41	42.8	0.85	44	120	230
GO01	6607	2.6%	59.5	38	70	1.18	31	165	199
LO04	6009	2.3%	45.5	35	29.7	0.65	29	110	139
MO04	5491	2.1%	68.5	68	37.1	0.54	47	134	417
GO17	5482	2.1%	70.2	67	44.6	0.64	47	150	279
GO18	5300	2.1%	70.5	63	43.5	0.62	48	151	287
GO06	5226	2.0%	82.4	47	82.2	1	60	265.8	169
LO05	5206	2.0%	52.8	43	35.2	0.67	42	122	221
LO01	5202	2.0%	53.2	47	28.3	0.53	28	104	257
GO05	4905	1.9%	103.9	77	78.5	0.76	77	246.8	309
KO06	4776	1.9%	58.5	50	37.7	0.64	39	129.2	55
GO07	4751	1.8%	76.6	66	51.6	0.67	45	170	316
MO03	4566	1.8%	78.9	64	57.1	0.72	48	198	388
MO06	4552	1.8%	52.3	44	33	0.63	33	117	333
GO04	4535	1.8%	102.5	72	87.1	0.85	68	277	270
MO05	4512	1.8%	71.3	64	41.7	0.58	59	146	413
GEG1	4498	1.8%	22	19	16	0.73	13	40.1	30
GO15	4327	1.7%	86.4	76	50.8	0.59	66	178.7	327
GO03	4204	1.6%	95.7	83	54.8	0.57	55	194	189
LO03	4196	1.6%	68.4	59	39.8	0.58	61	142	280

ActualOR	Count	Percent	Mean	Median	SD	CV	IQR	P95	UniqueProc
GO16	4100	1.6%	92.6	96	40.8	0.44	53	155	282
KO01	4074	1.6%	77.4	73	37.2	0.48	49	139	38
MO01	4063	1.6%	63.1	54	40.3	0.64	57	135	272
KO02	3923	1.5%	83.1	80	38.4	0.46	52	149	32
KO07	3651	1.4%	77.5	66	53.6	0.69	74	177.5	78
GO02	3514	1.4%	110.1	86	92.1	0.84	86	280.7	243
GO13	3308	1.3%	138.3	112	88.1	0.64	84.2	311	196
KO05	3024	1.2%	82.1	72	42.6	0.52	40	155	82
GO12	2889	1.1%	145.7	113	94.9	0.65	91	341	161
GOP1	2739	1.1%	38.3	37	20.3	0.53	20	70	51
GO09	2640	1.0%	143.6	119	96.9	0.67	128	337	263
GO08	2615	1.0%	93.6	63	76.7	0.82	71	262	163
GEE1	2330	0.9%	24.3	22.5	13.1	0.54	14	48	14
GSE1	2261	0.9%	23.8	21	14.7	0.62	16	50	19
GO10	1752	0.7%	283.8	300.5	146	0.51	190.2	497.9	180
GEE2	1646	0.6%	22.1	21	10.9	0.49	12	42	7
GOP2	1316	0.5%	37.9	35	21.8	0.58	24	77	65
GEX1	634	0.2%	30.1	26.5	22	0.73	14	55	9
ERCP	40	0.0%	40.6	37.5	20.7	0.51	18.8	69.4	5
GIC7	2	0.0%	159.5	159.5	29	0.18	20.5	178	2

The distribution of actual surgeries across operating rooms closely mirrors the planning data. A small number of rooms, including LP01, LP02, and MKI3, account for the largest share of total surgical activity, each exceeding 10000 to 20000 procedures. The remaining rooms show a gradual decline in volume, with most handling between 2000 and 6000 surgeries. Toward the end of the chart, several rooms record only a few hundred cases. This uneven distribution confirms that operating room usage is concentrated in a limited subset of rooms, while others are used less intensively or serve more specialized purposes. Figure 8 shows the full distribution.

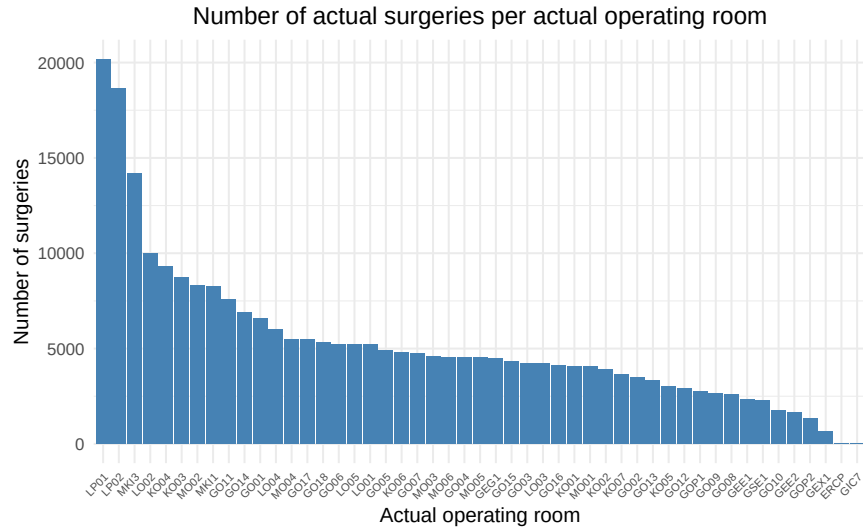


Figure 8: Number of actual surgeries per actual operating room

DurationMinutes

The average observed surgery duration is 55 minutes, with a median of 36 minutes, showing that most procedures are relatively short but a smaller group of long surgeries increases the mean. The interquartile range of 52 minutes indicates substantial variability across cases. Nearly all surgeries finish within 155 minutes, and only one percent exceed 302 minutes, while the longest recorded case lasts 1215 minutes. This confirms a strong right skewed distribution, where the majority of surgeries are short, and long duration cases represent rare but significant outliers. Table 10 summarises these statistics.

Table 10: Actual surgery duration (minutes), summary statistics

Mean	Median	SD	IQR	Min	Max	P95	P99	P99.9
55	36	58.5	52	1	1215	155	302	519

The distribution of actual surgery durations follows a similar right skewed pattern as the planned durations. Most surgeries last less than 60 minutes, and the frequency decreases steadily as duration increases. Only a small fraction of cases exceed 150 minutes, and very few extend beyond 250 minutes. This concentration of short procedures reflects the overall composition of surgical activity, where brief interventions dominate and long operations are relatively rare. The trimmed view at the 99th percentile, shown in Figure 9, helps visualise the main body of the data without distortion from extreme outliers.

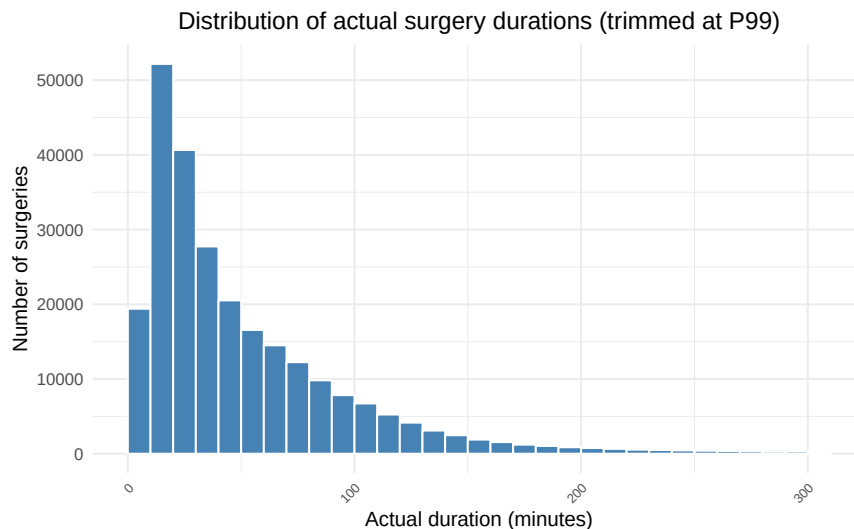


Figure 9: Distribution of actual surgery durations (trimmed at P99)

Deviation metrics

Deviation metrics capture the numerical differences between planned and actual performance at different points in the workflow. They quantify how accurately start times, end times, and total durations match the original plan. Analysing these deviations helps identify where delays or overruns of a particular surgery most often occur and whether they are systematic or random. This section provides the foundation for understanding the root causes of schedule inaccuracy.

StartDiff

The difference between actual and planned start times shows substantial variation across the dataset. On average, surgeries start 20.5 minutes later than planned, with a median delay of 9 minutes. However, the extremely high standard deviation reflects the presence of a few implausible or extreme values, which likely represent registration anomalies rather than genuine scheduling deviations. Table 11 summarises these deviation metrics.

In total, 28.9 percent of cases start more than 30 minutes late, while 13.1 percent begin more than 30 minutes early. This indicates that roughly one in three surgeries deviates noticeably from the planned start time, with delays being more common than early starts. These figures provide a first quantitative indication of how frequently the operating schedule diverges from the plan at the start of the day.

Table 11: Difference between actual and planned start time in minutes

Mean	Median	SD	IQR	Min	Max	P95	P99	P99_9	> 30 min late	< -30 min early
20.5	9	107.5	46	-1424	2255	120	546	1120	28.9%	13.1%

The distribution of start time deviations is centred around zero but shows a clear skew toward positive values, meaning that delays are more frequent than early starts. Most surgeries begin within ± 30 minutes of the planned time, but a significant share shows larger deviations, extending up to two hours in both directions. The sharp central peak indicates that a majority of cases start close to schedule, while the long tails confirm that extreme delays, although less common, occur regularly. Overall, the pattern suggests that while day to day planning is generally accurate, start time punctuality remains a challenge for a substantial portion of surgeries. Figure 10 shows the distribution grouped in ten minute intervals.

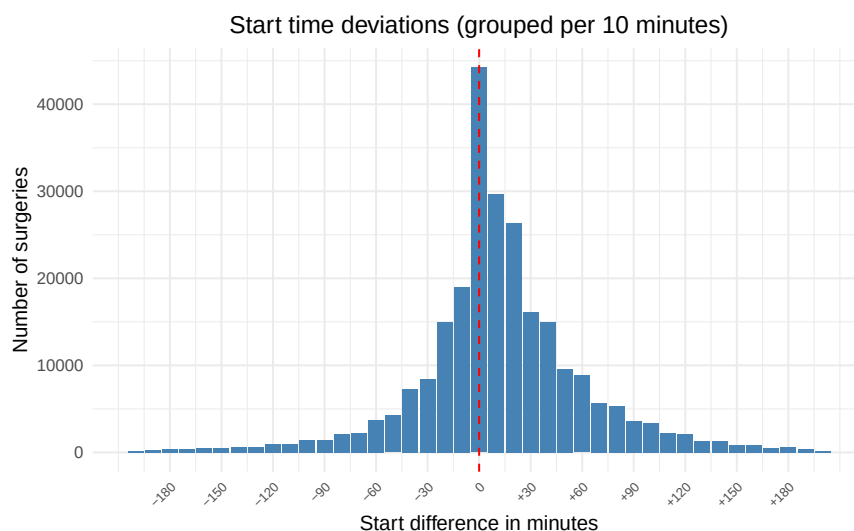


Figure 10: Start time deviations grouped per ten minutes

Among the one percent most extreme start time deviations, a small group of procedure types appears more frequently than others. Table 12 shows that CORO RADIAAL and PTCA HOSP stand out with 116 and 96 cases respectively, which together account for a considerable number of delayed or early starts simply due to their large overall presence in the dataset. Other procedures, including distal radius fractures, hip fracture treatments, caesarean sections, total hip replacements, extractions, and laparoscopic interventions, also

appear regularly among the outliers. This suggests that extreme deviations are not limited to a specific type of surgery but occur across a diverse range of activities, with frequency mainly reflecting overall procedure volume.

Table 12: Top 10 procedure types among the one percent most extreme start time deviations

ProcedureName	n
CORO RADIAAL	116
PTCA HOSP	96
POLS FRAKTUUR DISTAAL RADIUS	92
HEUP FRACTUUR TFNa	84
SECTIO	69
HEUP / TOTALE HEUPPROTHESE	57
EXTRACTIE WHT	56
EXPLORATIEVE LAPARASCOPIE	52
LAP.APPENDIX	52
ENKEL FRAKTUUR AO KLEIN FRAGMENT	50

The analysis of operating rooms with the most extreme start time deviations shows that a small number of rooms account for a large share of the outliers. GO11 stands out clearly, with 1828 extreme cases, far exceeding any other room. Other frequently affected rooms include KO03 with 189 cases, KO06 with 113 cases, and KO04 with 61 cases. The remaining rooms show much smaller counts. This pattern suggests that start time deviations are not evenly distributed across the operating theatre complex but tend to cluster in a few high activity or logistically sensitive rooms. Table 13 provides an overview of the ten operating rooms most frequently involved.

Table 13: Top 10 operating rooms among the one percent most extreme start time deviations

ActualOR	n
GO11	1828
KO03	189
KO06	113
KO04	61
GO06	27
GO05	26
GO04	23
GO07	23
MO03	21
GO13	17

Among the most extreme one percent of start time deviations, the vast majority are late starts. A total of 2394 cases, or 92.9 percent, begin later than planned, while only 183 cases, or 7.1 percent, start earlier. This clear imbalance confirms that extreme deviations almost always occur in the direction of delay rather than early execution. It suggests that structural or operational factors are more likely to cause postponements than premature starts, making delayed openings the dominant source of major scheduling disruptions. Table 14 summarises this distribution.

Table 14: Distribution of direction, early versus late, among the one percent most extreme cases

Direction	n	Percentage
Early	183	7.1%
Late	2394	92.9%

EndDiff

The difference between actual and planned end times shows a clear tendency toward delayed finishes. On average, surgeries end 18.9 minutes later than planned, with a median delay of 10 minutes. About 31.3 percent of cases finish more than 30 minutes late, while 16.0 percent end more than 30 minutes early. The high standard deviation results from a few extreme timestamps that fall outside realistic operating times. These figures indicate that small delays are common, but very large deviations occur only in a limited number of cases. Table 15 summarises these results.

Table 15: Difference between actual and planned end time in minutes

Mean	Median	SD	IQR	Min	Max	P95	P99	P99_9	> 30 min late	< -30 min early
18.9	10	109.3	56	-2016	2310	128	504	1092	31.3%	16.0%

The distribution of end time deviations is centred around zero but slightly skewed toward positive values, meaning that delays are more frequent than early completions. Most cases finish within ± 30 minutes of the planned time, but the long right tail shows that some surgeries extend well beyond expectations. This pattern closely resembles the distribution observed for start times, confirming that once a delay occurs early in the day, it often carries through to the end of the procedure. Figure 11 shows the distribution grouped in ten minute intervals.

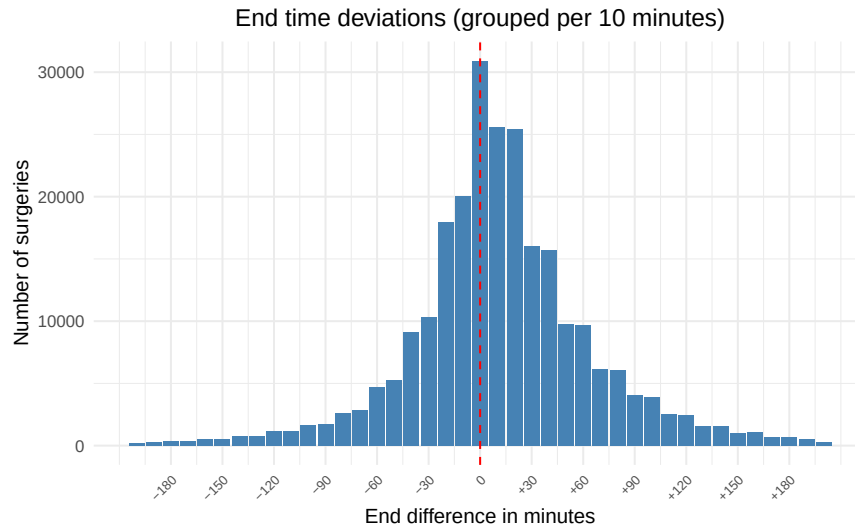


Figure 11: End time deviations grouped per ten minutes

Among the one percent most extreme end time deviations, procedures such as CORO RADIAAL, distal radius fracture repairs, and hip fractures (TFNa) appear most frequently, with 98, 93, and 87 cases respec-

tively. These are high volume procedure types, which explains their presence among the outliers. Other recurrent procedures include caesarean sections, extractions, laparoscopic interventions, total hip replacements, and ankle fracture treatments. This variety indicates that large deviations occur across different kinds of interventions rather than being specific to a single surgical category. Table 16 summarises the ten procedure types that appear most often among the extreme cases.

Table 16: Top 10 procedure types among the one percent most extreme end time deviations

ProcedureName	n
CORO RADIAAL	98
POLS FRAKTUUR DISTAAL RADIUS	93
HEUP FRACTUUR TFNa	87
SECTIO	71
EXTRACTIE WHT	66
HEUP / TOTALE HEUPPROTHESE	64
EXPLORATIEVE LAPARASCOPIE	57
LAP.APPENDIX	56
ENKEL FRAKTUUR AO KLEIN	51
FRAGMENT	
LAP.GALBLAAS	48

A small number of operating rooms account for most extreme end time deviations. GO11 clearly dominates, with 1957 cases, while KO03, KO06, and KO04 follow at a distance with 101, 98, and 52 cases respectively. The other rooms in the top ten register far lower numbers, mostly between 19 and 29 cases. This confirms that end time overruns are not evenly distributed but tend to cluster in a limited set of frequently used or high volume rooms. Table 17 provides an overview of the ten rooms most frequently involved.

Table 17: Top 10 operating rooms among the one percent most extreme end time deviations

ActualOR	n
GO11	1957
KO03	101
KO06	98
KO04	52
GO05	29
GO04	24
GO06	21
GO07	19
GO13	19
KO05	19

Among the most extreme end time deviations, 87.1 percent of cases end later than planned, compared to 12.9 percent that finish early. This consistent asymmetry confirms that schedule overruns are the dominant pattern, similar to what was observed for start times. Large early completions remain rare, suggesting that delays accumulate more easily than time savings during the surgical process. Table 18 summarises this distribution.

Table 18: Distribution of direction, early versus late, among the one percent most extreme end time deviations

Direction	n	Percentage
Early	332	12.9%
Late	2247	87.1%

DurationDiff

When distinguishing between longer and shorter cases, a clear asymmetry appears. About 42.7 percent of surgeries take longer than planned, with an average overrun of 16.4 minutes and a median of 9 minutes, while 57.3 percent finish sooner, with an average gain of 14.8 minutes and a median of 7 minutes. The standard deviation is much larger for shorter cases, 52.5 versus 22.2 minutes, indicating that a small number of procedures finish far earlier than planned, whereas most overruns remain more tightly clustered around their mean. Overall, however, the majority of operations stay close to their planned duration, confirming that duration estimates are generally reliable. Table 19 summarises these patterns.

Table 19: Difference between actual and planned duration, absolute minutes, separated by direction

direction	Mean	Median	SD	IQR	Min	Max	P95	P99	P99_9	Percent
Longer than planned	16.4	9	22.2	16	1	651	56	106	216.6	42.7%
Shorter than planned	14.8	7	52.5	12	0	1409	40	84	908	57.3%

Among the one percent most extreme duration deviations, the procedures with the highest number of outliers are dominated by high volume interventions. CORO RADIAAL appears most frequently with 157 extreme cases, of which 136 are shorter and 21 are longer than planned. COR.AORTA BYPASS GRAFT V. follows with 113 extreme deviations, consisting of 61 longer and 52 shorter cases. PTCA HOSP ranks third with 107 extreme observations, largely driven by shorter than planned durations. Other procedures such as RX interventionele neuro, P.L.I.F. MET BRAINLAB AERO, and exploratory laparoscopic or laparotomic surgeries also appear regularly among the outliers. The distribution shows that extreme deviations occur across a diverse set of procedures rather than being limited to a single surgical category. Table 20 provides the ten procedures most frequently involved.

Table 20: Top 10 procedure types with the highest number of extreme duration deviations

ProcedureName	Shorter	Longer	Total
CORO RADIAAL	136	21	157
COR.AORTA BYPASS GRAFT V.	52	61	113
PTCA HOSP	101	6	107
RX interventionele neuro	9	31	40
P.L.I.F. MET BRAINLAB AERO	14	25	39
EXPLORATIEVE LAPARASCOPIE	6	31	37
AVR	23	12	35
CABG OFF PUMP	24	3	27
EXPLORATIEVE LAPAROTOMIE	6	21	27
Recanalisatie intracranieel stroke	15	12	27

A limited number of operating rooms account for the majority of extreme duration deviations. GO10 stands out with 261 extreme cases, followed closely by KO03 with 256 cases. GO11, GO09, GO05, and GO04 also show substantial numbers, each with between 120 and 166 extreme observations. The remaining rooms in the top ten range between 98 and 137 cases. In several rooms, longer than planned procedures slightly outnumber shorter ones, while in others the opposite pattern appears, but in all cases the total number of large deviations is considerable. This confirms that the largest mismatches between planned and actual duration are concentrated in a small set of high activity rooms, many of which also recur in the analyses of start and end time deviations. Table 21 summarises these operating rooms.

Table 21: Top 10 operating rooms with the highest number of extreme duration deviations

ActualOR	Shorter	Longer	Total
GO10	131	130	261
KO03	222	34	256
GO11	43	123	166
GO09	60	77	137
GO05	53	80	133
MO03	25	96	121
GO04	64	56	120
GO12	50	65	115
KO07	32	75	107
KO06	33	65	98

Among the most extreme one percent of duration deviations, 1415 cases, or 55.0 percent, are longer than planned, while 1159 cases, or 45.0 percent, are shorter. This distribution shows a much smaller asymmetry than expected, indicating that extreme deviations occur in both directions. Table 22 summarises this distribution.

Table 22: Distribution of direction among the one percent most extreme duration deviations

Direction	n	Percentage
Longer	1415	55.0%
Shorter	1159	45.0%

percentage_deviation

When expressing the differences between planned and actual durations in relative terms, a clear asymmetry emerges between longer and shorter cases. Surgeries that last longer than planned deviate on average by 39.3 percent, with a median deviation of 25 percent, while procedures that finish sooner deviate by 23.8 percent on average and 20 percent at the median. This shows that overruns tend to be larger in magnitude than underruns, even though both occur frequently. The interquartile ranges of 38.9 percent for longer cases and 24.2 percent for shorter cases indicate substantial variability across cases, meaning that half of all surgeries differ from their expected duration by at least 20 to 25 percent. While most procedures stay within moderate limits, the upper percentiles, with P95 equal to 120 percent for longer cases and 60 percent for shorter cases, reveal that some operations still take considerably longer than scheduled, underscoring that planning accuracy varies strongly between procedures and specialties. Table 23 summarizes these patterns.

Table 23: Relative deviation, separated by direction

direction	Mean	Median	SD	IQR	Min	Max	P95	P99	P99.9
Shorter than planned	23.8	20	18.7	24.2	0	99.7	60	83.3	96.7
Longer than planned	39.3	25	67.8	38.9	0.2	13400	120	220.8	500

The distribution of percentage deviations is centred around zero, indicating that many surgeries finish close to their planned duration. Slightly more procedures finish earlier than planned, as reflected in the small shift of the peak toward negative values. Most surgeries deviate within about 40 percent of their planned time, while large over or underestimations are relatively uncommon. The narrow central peak suggests that planning is generally accurate, though the overall spread shows that notable deviations still occur in a substantial share of cases. Figure 12 displays the distribution grouped per ten percent.

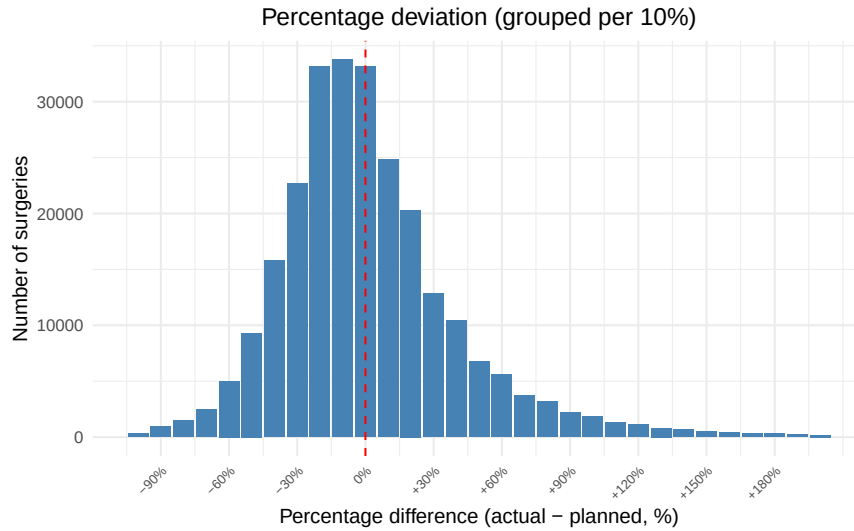


Figure 12: Percentage deviation (grouped per 10 percent)

The procedures with the highest average relative deviations show considerable differences between planned and actual durations. EXCISIE LIPOOM CHIR, MK has the largest mean deviation, followed by COLOR-POFRAFIE ANTERIOR and DVC, MK. For these procedures the average operating time is more than twice as long as planned. While the underlying causes have not been investigated in detail, these results indicate that for some procedure types the planned duration does not always reflect the time actually required in practice. Further analysis would be needed to determine whether this relates to scheduling conventions or other operational factors. Figure 13 shows the top twenty procedure types with the highest mean relative deviation.

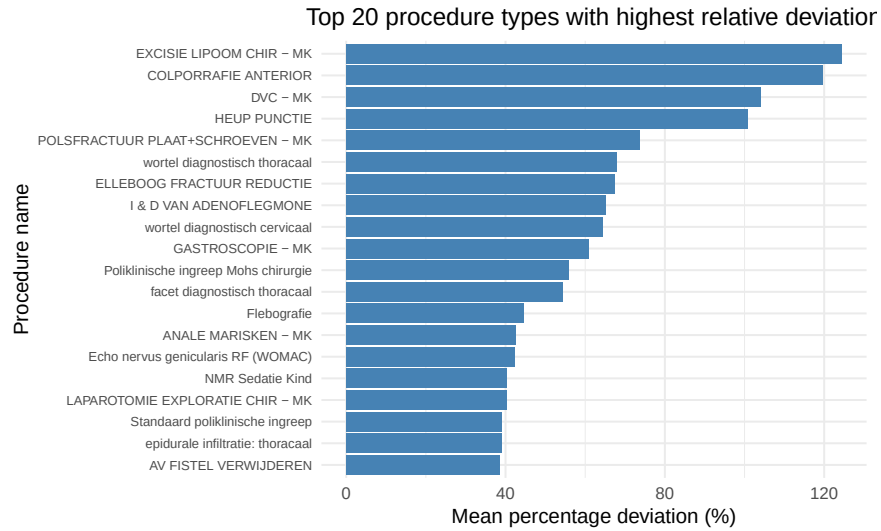


Figure 13: Top 20 procedure types with highest relative deviation

The average relative deviation varies noticeably by weekday. From Monday to Friday, mean deviations remain between roughly 2% and 4%, indicating consistent planning accuracy during regular operating days. In contrast, Saturday and Sunday show much higher deviations of approximately 20% and 21%. This pattern suggests that weekend cases are less predictable in duration, likely because they involve a different case mix than weekday surgeries. Figure 14 shows the mean relative deviation for each weekday.

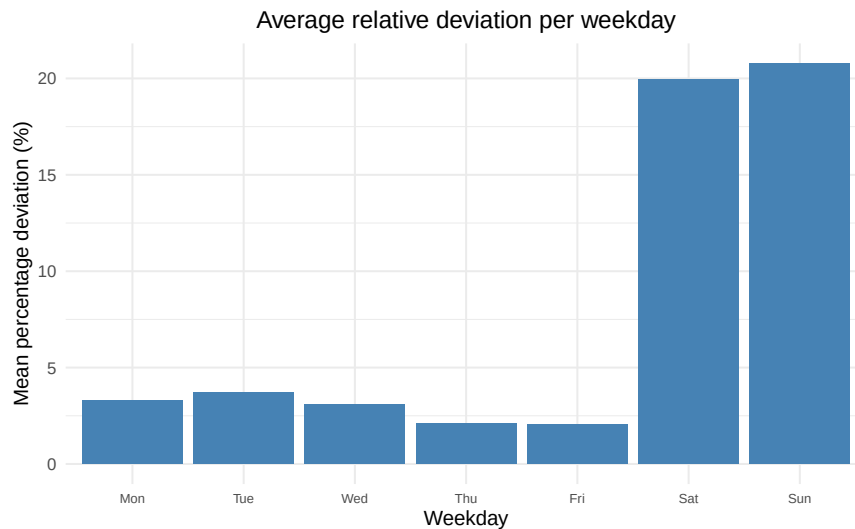


Figure 14: Average relative deviation per weekday

Conclusion

The exploratory analysis provides an overview of surgical activity at Ziekenhuis Oost Limburg across multiple dimensions of time, procedure type, and operating room planning. The results show that most surgeries are elective and performed during regular daytime hours, with short hospital stays and a strong predominance of day surgery. This reflects a high throughput surgical system mainly oriented toward planned interventions.

The descriptive statistics also show considerable variability in planned and observed metrics between operating rooms and procedure types. The wide range of planned durations, the uneven distribution of workload across rooms, and the long tail of hospital stays highlight the operational complexity behind the overall averages. These patterns underline the need for data driven performance metrics that go beyond simple utilization or volume counts.

The insights from this baseline analysis form the foundation for more advanced performance evaluation. By quantifying deviations between planned and observed activity, it becomes possible to identify where scheduling accuracy and resource use can be improved.